

INTRODUCTION

Similarly to other Slavic languages, Ukrainian has an extensive system of impersonal constructions. In the literature, the term *impersonal* has become an umbrella term for a variety of constructions, including those with semantically impersonal verbs and sentences that are treated as impersonal from the syntactic point of view (Méndez-Naya and López-Couso, 1997; Prenner, 2023). The former comprises a limited number of verbs denoting natural phenomena and physical and mental states. The latter is a large group of constructions built with verbs that are not inherently impersonal and can be used in personal sentences too.

For this project, I chose three types of Ukrainian impersonal constructions, including weather constructions and constructions with verbs describing physical states, -no/-to constructions and 3PL impersonals.

Slavic weather verbs are traditionally treated as truly subjectless (Hristov, 2023). While this contradicts the Subject Condition stating that every verbal predicate must have a subject, Ukrainian weather predicates have a zero-argument structure and can take neither overt nor covert subject. Some examples of such verbs include *sutenity* (to get dark), *vechority* (to get late), *doshchyty* (to rain), *snizhyty* (to snow), *mzhychyty* (to drizzle), *morozyty* (to freeze), etc.

Verbs denoting physical states belong to inherently impersonal predicates. They are never accompanied by a subject, but in contrast to weather verbs, take an accusative object. Verbs of physical state are even more infrequent than weather verbs and include *lykhomanyty* (to shake with fever), *nudyty* (to be nauseous), *sudomyty* (to spasm), *morozyty* (to have chills), etc.

-no/-to constructions get their name due to the invariant form of the predicate ending in *-no* or *-to*. In many ways, they resemble passive sentences but act unusually in terms of case assignment by preserving an accusative object. The Ukrainian -no/-to construction and its Polish counterpart have generated extensive debates over their status in both languages due to the fact that while being identical on the surface, they behave differently.

Lastly, in 3PL impersonal constructions, as the name suggests, the predicate always occurs in the third plural form. Sentences of this type can easily be transformed into personal sentences. These constructions lack a referential subject and the agent is perceived as indefinite (someone) or generalised (they, people).

The goal of this project is to implement a grammar that can provide an analysis of these three types of Ukrainian impersonals. The constructions are chosen based on the differences in the mechanics involved in creating the impersonality effect.

Ukrainian has a rich case marking system and a relatively flexible word order, with the subject-verb-object considered the neutral structure. Similarly to Russian, it is classified as a discourse-configurational language, where word order encodes discourse information. While the discourse level of analysis is out of scope of this work, I use insights from King's investigation into Russian (King, 1993) as well as general guidelines provided by Bresnan (Bresnan, 2001) and Dalrymple (2001) to propose a possible analysis for Ukrainian. As representatives of the East Slavic branch, Ukrainian and Russian share a lot of similarities, including the sentence structure. A thorough syntactic investigation may reveal more nuanced dissimilarities and justify a need for a completely different

approach to Ukrainian in contrast with Russian, but for the purpose of this work the analysis proposed by King appears to be sufficient. I also follow the LFG model introduced by Kibort and Maling (2015) and adopt the approach Kibort developed for analysing Polish impersonal constructions (Kibort 2004, 2006, 2008).

IMPERSONAL VERBS

The implementation of Ukrainian weather verbs is straightforward. They are considered true impersonal verbs (Shevelov, 1963) and never take an overt subject. Weather predicates fail the tests proposed by Kibort (2004) and Dalrymple (2001): they cannot be used in any number and person other than the default 3SG.NEUT and are incompatible with participial modifiers. This allows us to conclude that weather constructions do not have a covert subject either. Expletives like *it* and *there*, which act as dummy subjects in weather constructions in English, are not used in Ukrainian. Weather verbs are always intransitive and take neither an accusative nor a dative object.

To account for all these features, it is suggested to treat weather verbs as a special type of zero-argument predicates VERB-0ARG with an empty argument structure.

$$\text{VERB-0ARG(P)} = (\wedge \text{PRED}) = \text{'P<>'}$$

This solution allows the grammar to successfully analyse a sentence consisting of one predicate, such as *Svitaie*. (It dawns) and to provide an f-structure for it.

PRED	'svitaty'
AGR	[GEND neut, NUM sg, PERS 3]
TNS-ASP	[MOOD indicative, TENSE pres]
1[VFORM	fin

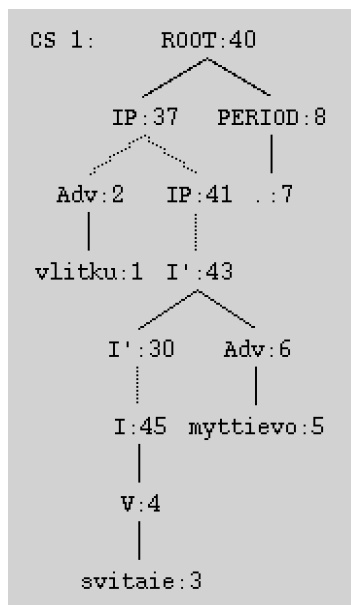
Without a subject that normally triggers the agreement, weather predicates only carry the default agreement features including the NEUT gender, SG number and 3 PERS. As these features could not be attributed to any specific grammatical function, I used the agreement feature structure AGR proposed by Alsina and Vigo (2017).

The grammar provides an accurate analysis for sentences with multiple adjuncts, such as *Vlitku svitaie myttievo*. (In summer it dawns instantly).

PRED	'svitaty'												
ADJ	<table> <tr> <td>1</td><td> <table> <tr> <td>PRED</td><td>'vlitku'</td></tr> <tr> <td>1[ADV-TYPE</td><td>sadv</td></tr> </table> </td></tr> <tr> <td>5</td><td> <table> <tr> <td>PRED</td><td>'myttievo'</td></tr> <tr> <td>5[ADV-TYPE</td><td>vadv</td></tr> </table> </td></tr> </table>	1	<table> <tr> <td>PRED</td><td>'vlitku'</td></tr> <tr> <td>1[ADV-TYPE</td><td>sadv</td></tr> </table>	PRED	'vlitku'	1[ADV-TYPE	sadv	5	<table> <tr> <td>PRED</td><td>'myttievo'</td></tr> <tr> <td>5[ADV-TYPE</td><td>vadv</td></tr> </table>	PRED	'myttievo'	5[ADV-TYPE	vadv
1	<table> <tr> <td>PRED</td><td>'vlitku'</td></tr> <tr> <td>1[ADV-TYPE</td><td>sadv</td></tr> </table>	PRED	'vlitku'	1[ADV-TYPE	sadv								
PRED	'vlitku'												
1[ADV-TYPE	sadv												
5	<table> <tr> <td>PRED</td><td>'myttievo'</td></tr> <tr> <td>5[ADV-TYPE</td><td>vadv</td></tr> </table>	PRED	'myttievo'	5[ADV-TYPE	vadv								
PRED	'myttievo'												
5[ADV-TYPE	vadv												
AGR	[GEND neut, NUM sg, PERS 3]												
TNS-ASP	[MOOD indicative, TENSE pres]												
3[VFORM	fin												

In the c-structure, in line with King's (1993) analysis, the finite verb *svitaie*, which carries tense and agreement features, is placed in the I node. Adjuncts are attached on different levels as one of them is of *sadv* type and modifies the entire clause and the other one is of *vadv* type and modifies the main verb. The

distinction is inspired by the English grammar on the INESS platform (Rosén, Smedt, Meurer, Dyvik, 2012).



For the sentence **Nebo svitaie*. (The sky is dawning), a well-formed f-structure cannot be built as it would violate the Principle of Coherence, stating that every grammatical function must be subcategorised by the predicate (Bresnan, 2001). The nominative word *nebo* can only occupy the SUBJ position, which in this case is not subcategorised by the verb. As expected, the grammar does not provide an analysis for this sentence.

Another subgroup of semantically impersonal verbs includes verbs of physical state. Sentences with predicates of physical state cannot have an overt subject and, similarly to weather predicates, do not pass subjecthood tests, which suggests that they lack a covert subject. At the same time, these verbs always require an accusative object. It can be assumed that they have a one-place argument structure, realised in the template VERB-OBJ.

$$\text{VERB-OBJ(P)} = (\wedge \text{PRED}) = \text{'P} < (\wedge \text{OBJ}) > \text{'}$$

PRED	'nudyty<[1:ya]>'
OBJ	1[PRED 'ya']
AGR	[CASE acc, NUM sg, PERS 1, PRON-TYPE pers]
TNS-ASP	[GEND neut, NUM sg, PERS 3]
3VFORM	[TENSE pres]
	fin

The default agreement is always 3SG neuter. The agreement features are combined in the AGR block. As intended, the grammar cannot provide an f-structure for the ungrammatical sentence with **Vas morozyt zastuda* (A cold is giving you chills.) due to the absence of the SUBJ function in the verb subcategorisation frame and case constraints put on the OBJ.

-NO/-TO CONSTRUCTIONS

While the -no/-to constructions received the most attention among all impersonal sentences in Ukrainian, there is no definite answer to how they should be analysed.

On the one hand, they share similarities with true passives. Both constructions only allow transitive predicates. In both, the agent from the corresponding active personal sentence can be expressed in the instrumental case. On the other hand, the -no/-to constructions never allow a nominative subject and retain an accusative object.

I follow the model proposed by Kibort and Maling (2015) who treat Ukrainian -no/-to constructions as a non-promotional passive in which the direct object is not promoted and the subject argument is not expressed. The analysis is based on the comparison with a similar Polish construction, which also lacks an overt subject but passes syntactic control and binding tests, proving there is a fully operational covert subject. The Ukrainian construction, on the other hand, does not pass these tests. The suggestion that -no/-to constructions should have a passive rather than a covert *pro* reading as in Polish is supported by many researchers (Lavine, 2005; Sobin, 1985; Shevelov, 1963).

Kibort and Maling's non-promotional passive model is implemented in XLE via extending the standard passive rule with an additional branch. This allows the grammar to suppress the subject or demote it to the OBL-AG while leaving the accusative object intact. The construction is identified as passive via assigning + value to the PASSIVE feature.

The constraining equation ($\wedge$ VFORM) =c no_to verifies that only -no/-to forms can appear in the construction. This category is not a part of the standard ParGram feature set, but it is introduced for the -no/-to forms for a number of reasons. These forms are invariant in modern Ukrainian and believed to be the passive past participle frozen in the neuter gender (Shevelov, 1963). They can only occur in this specific construction and cannot be classified as past participles, which are expected to agree with the subject in number and gender. The VFORM is also assigned to the auxiliary *bulo* as this is the only form of auxiliary that can occur with -no/-to predicates in the past tense. In true passive sentences, auxiliaries have to agree with the subject in number and gender.

The grammar accurately analyses -no/-to constructions with an accusative object in different genders and numbers. It also provides an accurate analysis for sentences with the OBL-AG in the instrumental case, as in *Ukaz bulo pidpysano prezidentom*. (The decree was signed by the president).

```
[
  PRED 'pidpysaty<[7:prezydent], [1:ukaz]>'
  OBJ 1 [
    PRED 'ukaz'
    CASE acc, GEND masc, NUM sg, PERS 3
  ]
  OBL-AG 7 [
    PRED 'prezydent'
    ANIMATE +, CASE inst, GEND masc, NUM sg, PERS 3
  ]
  TNS-ASP [MOOD indicative, TENSE past]
  3 [PASSIVE +, VFORM no_to]
]
```

The grammar differentiates between true passive and -no/-to constructions and provides correct f-structures for both sentences below.

Valizu bulo znaideno.
suitcase.ACC.SG.FEM be.PST.3SG.NEUT.AUX find.PASS.IMPERS
'The suitcase was found.'

```
[
  PRED 'znaity<NULL, [1:valiza]>'
  OBJ 1 [
    PRED 'valiza'
    CASE acc, GEND fem, NUM sg, PERS 3
  ]
  TNS-ASP [MOOD indicative, TENSE past]
  3 [PASSIVE +, VFORM no_to]
]
```

Valiza bula znaidena.
suitcase.NOM.SG.FEM *be.PST.3SG.FEM.AUX* *find.PTCP.NOM.SG.FEM*
 ‘The suitcase was found.’

```
[
  PRED   'znaity<NULL, [1:valiza]>'
  SUBJ   1 [
    PRED 'valiza'
    CASE nom, GEND fem, NUM sg, PERS 3
  ]
  TNS-ASP [MOOD indicative, TENSE past]
  3 [PARTICIPLE past, PASSIVE +
]
```

-no/-to constructions occur with the auxiliary *buty/bulo* (be) in the future and past tenses while in the present tense, auxiliaries are omitted. The form -no/-to lost its temporal meaning in Ukrainian (Shevelov, 1963) and does not carry tense information itself. To ensure that the f-structure contains tense information even if the auxiliary (or finite) verb is absent in the present tense, as suggested by ParGram guidelines, I add an additional *Inull* rule with an epsilon category to pass the *pres* value without an overt element appearing in the c-structure.

Valizu znaideno.
suitcase.ACC.SG.FEM *find.PASS.IMPERS*

```
[
  PRED   'znaity<NULL, [1:valiza]>'
  OBJ    1 [
    PRED 'valiza'
    CASE acc, GEND fem, NUM sg, PERS 3
  ]
  TNS-ASP [MOOD indicative, TENSE pres]
  3 [PASSIVE +, VFORM no_to
]
```

As expected, the grammar rejects -no/-to constructions with a noun in the nominative case or a past participle form instead of the -no/-to form.

3PL CONSTRUCTIONS

3PL constructions are also known as relatively impersonal. They lack an overt subject, but the agent of the action is perceived as an indefinite person or group of people. Researchers emphasize that the activity denoted by the predicate is associated with humans (Siewierska & Papastathi, 2011; Prenner, 2023; Paducheva, 2012; Bunčić, 2020). Paducheva (2012) notes that in 3PL constructions native speakers always interpret the action as performed by a human being even if the verb itself denotes an animal activity as in *Za oknom myaukali*. (Outside the window, [they] were meowing).

3PL constructions can be formed with different verbs, including transitive and intransitive predicates, both unergative and unaccusative. They do not take a nominative subject, and unlike -no/-to constructions, cannot have an instrumental agent.

From the syntactic point of view, 3PL impersonals behave as sentences with an active subject. The covert subject passes the tests introduced by Kibort (2008), including the subject-control, participial control and volition tests and reflexive pronoun binding.

The main challenge in modeling 3PL impersonals with XLE is that they are identical to regular 3PL sentences with anaphoric reference, as Ukrainian is a pro-drop language. But as Shevelov (1963) notes in 3PL impersonal sentences the implicit subject is generalised and can be replaced with the word *liudy* (people). The subject therefore can be interpreted as arbitrary.

The proposed analysis is implemented via the 3PL-IMPERS template, containing a set of SUBJ features. They include the PRON-TYPE feature with the *arb* value and the HUMAN feature with the + value. In line with the previous research, it is assumed that this specific construction receives a human interpretation. Normally Ukrainian does not distinguish between human and nonhuman nouns but rather between animate and inanimate.

```
[
  PRED 'perekryty<[5-SUBJ:pro], [3:vulytsia]>'
  SUBJ [
    PRED 'pro'
    HUMAN +, NUM pl, PERS 3, PRON-TYPE arb
  ]
  OBJ [
    PRED 'vulytsia'
    ADJ [
      PRED 'tsestralnyi'
      1[CASE acc, GEND fem, NUM sg, PERS 3]
    ]
    3[CASE acc, GEND fem, NUM sg, PERS 3]
  ]
  TNS-ASP [MOOD indicative, TENSE fut]
  5[PASSIVE -, VFORM fin]
]
```

If the nominative subject like *protestuvalnyky* (protesters) appears in this sentence, it stays grammatical but becomes personal.

```
[
  PRED 'perekryty<[1:protestuvalnyk], [5:vulytsia]>'
  SUBJ 1[
    PRED 'protestuvalnyk'
    ANIMATE +, CASE nom, NUM pl, PERS 3
  ]
  OBJ 5[
    PRED 'vulytsia'
    CASE acc, GEND fem, NUM sg, PERS 3
  ]
  TNS-ASP [MOOD indicative, TENSE fut]
  3[PASSIVE -, VFORM fin]
]
```

As expected, the grammar does not provide a *pro arb* analysis for clauses with predicates in person or number other than 3PL.

CONCLUSION

The main challenge in developing a Ukrainian grammar in XLE is the relatively free word order, which makes it necessary to extend the rules to allow multiple structures and provide an accurate analysis for all of them. Secondly, the pro-drop nature of the language leads to ambiguous analyses for some constructions, which can be resolved through context rather than syntactic features. For instance, 3PL impersonals from the examples analysed in this work, while being generalised in nature and not implying a specific agent, can be analysed as instances of pro-drop if they are preceded by a sentence with a possible referent. Overall, Ukrainian offers rich material for research and grammar modelling, including numerous other impersonal constructions.

REFERENCES

- Alsina, A., & Vigo, E. M. (2017). Agreement: Interactions with case and raising. In M. Butt & T. H. King (Eds.), *Proceedings of the LFG'17 Conference, University of Konstanz* (pp. 3–23). CSLI Publications. <http://csli-publications.stanford.edu/LFG/2017>
- Bresnan, J. (2001). *Lexical-functional syntax*. Blackwell Publishers.
- Bunčić, D. (2020). Agent prominence and movement in Russian third person plural impersonals (“indefinite-personal sentences”). *Die Welt der Slaven*, 65(2), 297–311. <https://doi.org/10.13173/WS.65.2.297>
- Dalrymple, M. (2001). *Lexical functional grammar* (Syntax and semantics, Vol. 34). Academic Press.
- Hristov, B. (2023). LFG and Slavic languages. In M. Dalrymple (Ed.), *Handbook of lexical functional grammar* (pp. 1753–1804). Language Science Press. 10.5281/zenodo.10037797
- Kibort, A. (2004). *Passive and passive-like constructions in English and Polish* [Doctoral dissertation, University of Cambridge].
- Kibort, A. (2006). On three different types of subjectlessness and how to model them in LFG. In M. Butt & T. H. King (Eds.), *Proceedings of the LFG06 conference, University of Konstanz* (pp. 289–309). CSLI Publications.
- Kibort, A. (2008). Impersonals in Polish: an LFG perspective. *Transactions of the Philological Society*, 106(2), 246–289. <https://doi.org/10.1111/j.1467-968X.2008.00213.x>
- Kibort, A., & Maling, J. (2015). Modelling the syntactic ambiguity of the active vs. passive impersonal in LFG. In M. Butt & T. H. King (Eds.), *Proceedings of the LFG15 Conference* (pp. 145–165). CSLI Publications.
- King, T. H. (1993). *Configuring topic and focus in Russian* [Doctoral dissertation, Stanford University].
- Lavine, J. E. (2005). The morphosyntax of Polish and Ukrainian -no/-to. *Journal of Slavic Linguistics*, 13(1), 75–117. <http://www.jstor.org/stable/24599548>
- López-Couso, M. J., & Méndez-Naya, B. (1997). What is really meant by impersonal?: On impersonal and related terms. *Atlantis: Revista de la Asociación Española de Estudios Anglo-Norteamericanos*, 19(2), 185–192.
- Paducheva, E. V. (2012). Neopredelenno-lichnoe predloženie i ego podrazumevaemyj subekt [The indefinite-personal sentence and its implied subject]. *Voprosy Jazykoznanija*, (1), 27–41.
- Prenner, M. K. (2023). *Agentivity in human impersonal constructions in Polish and Russian*. Peter Lang. 10.3726/b20385
- Rosén, V., De Smedt, K., Meurer, P., & Dyvik, H. (2012, May). An open infrastructure for advanced treebanking. In J. Hajič, K. De Smedt, M. Tadić, & A. Branco (Eds.), *Proceedings of the META-RESEARCH Workshop on Advanced Treebanking at LREC2012* (pp. 22–29). Istanbul, Turkey. <https://clarino.uib.no/iness>
- Shevelov, G. Y. (1963) *The syntax of modern literary Ukrainian*. Mouton.
- Siewierska, A., & Papastathi, M. (2011). Towards a typology of third person plural impersonals. *Linguistics*, 49(3), 575–610. <https://doi.org/10.1515/ling.2011.018>
- Sobin, N. J. (1985). Case assignment in Ukrainian morphological passive constructions. *Linguistic Inquiry*, 16(4), 649–662. <http://www.jstor.org/stable/4178461>

APPENDIX

1. Divchyna chytaie tsikavu knyzhku.
girl.NOM.SG.FEM read.PRS.3SG interesting.ACC.SG.FEM.ADJ book.ACC.SG.FEM
‘The girl is reading an interesting book.’
2. Khlopets mitsno spyt.
boy.NOM.SG.MASC soundly.ADV sleep.PRS.3SG
‘The boy is sleeping soundly.’
3. Vrantsi doshchylo.
in_morning.ADV rain.PST.3SG.NEUT
‘It rained in the morning.’
4. Vlitku svitaie myttievo.
in_summer.ADV dawn.PRS.3SG instantly.ADV
‘In summer it dawns instantly.’
5. *Nebo svitaie.
sky.NOM.SG.NEUT dawn.PRS.3SG
(intended) ‘The sky is dawning.’
6. Mene nudyt.
I.ACC.1SG.PRON nauseate.PRS.3SG
‘I feel nauseous.’
7. Dytynu lykhomanylo.
child.ACC.SG.NEUT fever.PST.3SG.NEUT
‘The child was feverish.’
8. *Vas morozyt zastuda.
you.ACC.2PL chill.PST.3SG cold.NOM.SG.FEM
(intended) ‘A cold is giving you chills.’
9. Nadvori znaideno valizu.
outside.ADV find.PASS.IMPERS suitcase.ACC.SG.FEM
‘A suitcase was found outside.’
10. Valizu bulo znaideno.
suitcase.ACC.SG.FEM be.PST.3SG.NEUT.AUX find.PASS.IMPERS
‘The suitcase was found.’
11. Valiza bula znaidena.
suitcase.NOM.SG.FEM be.PST.3SG.FEM.AUX find.PTCP.NOM.SG.FEM
‘The suitcase was found.’
12. Syniu sukniu poshyto kravchyneiu.
blue.ACC.SG.FEM dress.ACC.SG.FEM sew.PASS.IMPERS tailor.INST.SG.FEM
‘The blue dress was sewn by a tailor.’
13. Ukaz bulo pidpysano prezidentom.

decree.ACC.SG.MASC be.PST.SG.NEUT.AUX sign.PASS.IMPERS president.INST.SG.MASC

‘The decree was signed by the president.’

14. Kartyny bulo vykradeno.

picture.ACC.PL be.PST.SG.NEUT.AUX steal.PASS.IMPERS

‘The paintings were stolen.’

15. *Suknia poshyto kravchyneiu.

dress.NOM.SG.FEM sewn.PASS.IMPERS tailor.INST.SG.FEM

(intended) ‘The dress was sewn by a tailor.’

16. *Kartyny bulo vykradeni.

picture.ACC.PL be.PST.SG.NEUT.AUX steal.PTCP.ACC.PL.FEM

(intended) ‘The paintings were stolen.’

17. Vdoma vymknuly svitlo.

turn_off.PST.3PL turn_off.PST.3PL light.ACC.SG.NEUT

‘[They] turned off the light at home.’

18. Tsentralnu vulytsiu perekryiut.

central.ACC.SG.FEM.ADJ street.ACC.SG.FEM block.PST.3PL

‘[They] will block the central street.’

19. Vden tut zasmahaiut.

in_daytime.ADV here.ADV sunbathe.PRS.3PL

‘[They] sunbathe here during the day.’

20. Protestuvalnyky perekryiut vulytsiu.

protester.NOM.3PL turn_off.PST.3PL street.ACC.SG.FEM

‘Protesters will block the street’.

21. *Vdoma vymknula svitlo.

at_home.ADV turn_off.PST.3PL light.ACC.SG.NEUT

(intended) ‘At home, [she] turned off the light’.